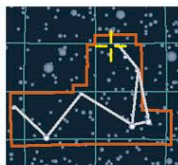


DARK NEBULA

Cone Nebula



CATALOG NUMBER
NGC 2264

DISTANCE FROM SUN
2,500 light-years

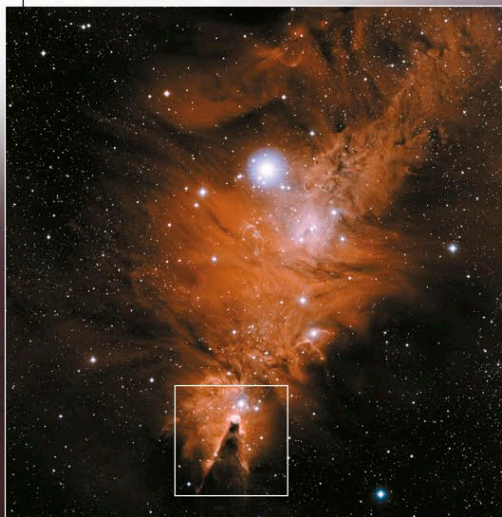
MAGNITUDE 3.9

MONOCEROS

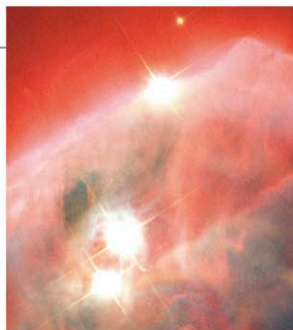
Discovered by William Herschel in 1785, the Cone Nebula is a dark nebula located at the edge of an immense, turbulent star-forming region. This conical pillar of dust and gas is more than 7 light-years long and at its "top" is 2.5 light-years across. The Cone Nebula is closely associated with the star cluster NGC 2264, commonly known as the Christmas Tree Cluster. This cluster,

CHRISTMAS TREE CLUSTER

The stars of the open cluster NGC 2264 can be seen in this image resembling an upside-down Christmas tree, with the Core Nebula (boxed) at the apex of the tree.



which spans a distance of 50 light-years, is made up of at least 250 stars, and it is the light from some of its newborn stars that allows the Cone Nebula to be seen in silhouette. The Cone Nebula is located at the top of the Christmas Tree Cluster, pointing downward to the bottom of the tree. At the opposite end, the 5th-magnitude star S Mon marks the left of the base of the tree (see below left). Jets of stellar material thrown out by newly forming stars have been detected within the star cluster. These Herbig-Haro objects also help shape the material in the surrounding nebula. One explanation for the shape of the Cone Nebula suggests that it was formed by stellar wind particles from an energetic source blowing past a Bok globule at the head of the cone. Buried in the dust and gas near



INFRARED IMAGING

Unseen in an optical image (left), a remarkable infrared view of the tip of the Cone Nebula (right) reveals, to the right of the image, a clutch of faint newborn stars.



the top of the Cone is a massive star known as NGC 2264 IRS, which is surrounded by six smaller Sun-like stars. It is thought that the outflow of stellar material during the early years of this massive star triggered the formation of the surrounding six and also helped to sculpt

the shape of the Cone Nebula itself. None of these stars are visible with optical telescopes. Infrared observations have revealed further embryonic stars embedded in the nebulosity (above), making this one of the most active star-forming regions in this area of the Milky Way.

TOWER OF RESISTANCE

Born in immense clouds of dust and gas, the great tower of the Cone Nebula is a slightly denser region of material that has resisted erosion by radiation from its neighboring stars.